Finite Spectral Shadows for the Collatz Valuation Cocycle:
Computational Companion

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Abstract

This companion documents the C numerical suite `collatz_tests` and the nine experiments (A–I) supporting the main paper, *Finite Spectral Shadows for the Collatz Valuation Cocycle* [3]. It reports the suite architecture, the exact construction of the finite residue–height operators and their Arnoldi spectral solver, the per-experiment pre-registrations and results, and a dedicated analysis relating the empirically detected "shadow" windows to the operator gaps. All figures are reproducible from the recorded runs. Section and equation references of the form "main, §X" point to the main paper. Throughout, the goal is diagnostic: to test whether the deterministic defect cocycle $W_{n,\ell}(x;\chi,s)$ is the right finite detector and where it sees structure.

Contents

1 The suite

`collatz_tests` is an ISO C11 program over the accelerated odd step $S(x) = (3x+1)/2^b$, $b = v_2(3x+1)$. Build profiles: FAST128 (`__uint128_t`, overflow-flagged), GMP (`mpz_t` [2]; large seeds, exact congruence checks), DEBUG (invariant assertions: $3x+1 = 2^b x'$, odd output, $b \geq 1$). Modules: `bigval` (the two backends), `orbit_scan` (streaming), `modarith`, `characters` (additive $\chi_k(a) = e(ka/Q)$ and multiplicative on $(\mathbb{Z}/3^r)^\times$), `defect_sums` ($W_{n,\ell}$ via complex prefix sums), `entropy_stats`, `critical_events` (excursions, segment labels), `residue_height_graph` (operator builder), `sparse_matrix` (CSR + Arnoldi), `io_csv`, `config`.

Design requirements (adopted from review). (1) the $\bmod 3^r$ congruence (main, Lemma 1) is a correctness *gate* (Exp. H), verified before any spectral code relies on it; (2) annealed and exact spectral radii are reported in *separate* columns and never substituted (Exp. F); (3) the shadow search reports per-window magnitude *and* length so both lim sup plateaus and long sparse blocks are visible (Exp. I); (4) the quenched pressure is computed as $\limsup_n \frac{1}{n}\sum s\Delta_{b_i}$, not an ill-defined sum over words.

Provisional criticality. The project-internal "C_A-critical" certificate is not fixed; the suite offers `--crit-mode` with `vsq` (the literal $V^2 \leq 2^A P$, which is degenerate) and `height` (peak $\geq$ launch$\cdot 2^A$, the default). **All conclusions tagged `critical_C_A`, `high_ascent`, `post_exit` (Exp. B, D, I) are exploratory until that certificate is fixed.**

2 Operators and the spectral solver

The finite operator $\mathcal{L}_{s,Q,L,\chi}$ acts on states (a,z), $a \in U_Q$, $z \in \{0,\dots,L\}$, with the source-twist convention of the main paper. The residue map is forward, $a' = (3a+1)2^{-b} \bmod Q$ for odd Q (2 invertible), and inverse-branch $a' = 3^{-1}(2^b a - 1)$ for Q coprime to 3; the height updates $z' = z + \lfloor R(\log_2 3 - b)\rceil$ with killing off $[0, L]$.

Height coupling. `--height-mode sync` (default, faithful) uses one valuation b for both residue and height; `factored` builds $\mathcal{L}_{\mathrm{res}}(\chi) \otimes T_{\mathrm{height}}$ with an independent height valuation. The main paper shows `sync` is the correct vehicle and `factored` is a diagnostic only.

Arnoldi solver. Power iteration fails on the killed-height operator (a near-degenerate top cluster from the height-transport mode: residuals 3–9%, and spurious gap ratios > 1 violating Wielandt). It is replaced by a restarted complex Arnoldi [1, 4]: complex Givens rotations, a Hessenberg single-shift QR eigensolver (Wilkinson shift, deflation), modified Gram–Schmidt with one reorthogonalisation, Krylov dimension grown to convergence. Validated against known spectra (real/complex diagonal, cyclic permutation = roots of unity, agreement with power iteration). All operator runs then converge to residual $< 8 \times 10^{-11}$ with no Wielandt violations.

3 Experiments A–I

H — congruence gate

Verifies main, Lemma 1.

Verified

23,905,968/23,905,968 checks match (FAST128, seeds 3–10^6, $r \leq 8$, 0 overflow) and 144/144 under GMP for a 16-digit seed at $r = 12$, $n \leq 155$ (values $\gg 2^{128}$).

A — height-tail & critical census

Pre-registered: $\log_2 \Pr(h \geq H) \approx -H + C$. Octave 24, 10^5 seeds: the per-step slope over $H = 3, \ldots, 11$ averages ≈ -0.97. **Confirmed.**

B — empirical Cramér-tilt law

Pre-registered: critical-excursion bulk $\to p_b^{(1)} = 3 \cdot 4^{-b}$, entropy/digit $\to H_2(3/4) = 0.8113$.

Direction confirmed (provisional labels)

Octave 26, 2×10^5 seeds, boundary-excluded *middle* segments (1.0×10^5 segs): $\hat{p} = (0.657, 0.229, 0.089, 0.021)$, $D\big(\hat{p} \,\|\, p^{(1)}\big) = 0.093 \ll D(\hat{p} \,\|\, p) = 0.191$, entropy/digit 0.870.

C — defect windows

$|W|$ at $s = 0$ decreases with window length (0.31 at $L{=}16 \to 0.11$ at $L{\approx}116$, $Q = 27$). Single-orbit windows are short (orbits terminate), so the asymptotic decorrelation is read from the operators (E/F), not single orbits.

D — post-exit taboo (provisional)

6,496 post-exit windows: $D(\hat{p}_{\text{post}} \,\|\, p) = 0.177 \ll D\big(\hat{p}_{\text{post}} \,\big\|\, p^{(1)}\big) = 0.576$, mean $M_{\text{post}}(1) = 0.989 < 1$. Post-exit relaxes toward annealed, away from the critical tilt. (Conditional on the working criticality definition.)

E/F — finite operator gaps

Gaps present; exact $\neq$ annealed; constant by conductor

Residue-only: every nontrivial sector gapped; the χ_2 ratio equals the closed form $\frac{2^{1+s}-1}{2^{1+s}+1}$ (main, Prop.) to the digit. Exact vs annealed Fourier (48 sectors): corr $= 0.38$, mean deviation 0.25—the exact gap is *not* the i.i.d. value. On the faithful `sync` operator after the coboundary quotient, the gap ratio is r-independent and organised by conductor (main, Prop. on autonomy):

conductor	$3\,(\chi_2)$	9	27	81	243
order	2	6	9	54	81
gap ratio (s=0)	1.0000	**0.9069**	0.8530	0.8261	0.7306

each entry equal across $Q \in \{9, 27, 81, 243\}$; worst non-coboundary sector at the fixed conductor 9, $\kappa_0 \approx 0.0931$.

G — trace/determinant shadows

Small (Q, L): $\text{tr}(\mathcal{L}^1) = 0$, $\text{tr}(\mathcal{L}^2) = 1.25$, $\text{tr}(\mathcal{L}^4) = 0.594$; $\det(1 - z\mathcal{L})$ smooth and positive—no spurious peripheral structure.

I — quenched spectral-shadow search (provisional labels)

Shadow flag: $\max_{Q,\chi,s} |W| \geq \delta$ *and* $\geq 3\times$ a length-scaled null. Over 1,254,232 windows, 4.2% flag; by label `critical_C_A` 8.4% > `high_ascent` 5.4% > `post_exit` 3.2%; shadow rate grows with length to 100% at length ≥ 64 ($n = 218$).

4 Relating shadows to operator gaps

We join the flagged shadow windows (moduli $\{9, 27, 81, 243\}$, all powers of 3, multiplicative characters) to the residue-only operator gaps (Exp. F) on (Q, χ-index, s).

Tilt-level correspondence (clean); sector-level confounded

Of 53,158 shadow windows, 82% (and 100% of the 218 long ones) sit at the critical tilt $s = 1$, where F's gaps are weakest; long shadows sit in weaker-gap sectors (mean gap ratio 0.73 vs. 0.57). But $\mathrm{corr}(|W|, \mathrm{gap}) = +0.25$ only: 94% of shadows pick the smallest modulus $Q = 9$ (six unit residues), so the per-sector signal is dominated by finite residue-class inflation, not the asymptotic gap.

Interpretation. The finite operator decorrelates every non-coboundary sector; the long shadows track it at the tilt level and in gap magnitude, but the per-orbit signal is confounded by small-Q finite size—the circularity leak made explicit. This reinforces that the operator gap (not the single-orbit Weyl sum) is the reliable object, and that the faithful `sync` operator on the coboundary quotient is the right vehicle (main, §Refined target).

5 Reproducibility

Every output is append-only CSV with a provenance header (git commit, compiler, build profile, command line, UTC timestamp). The large-scale sweep is regenerated by a single driver script; the operator runs use the Arnoldi solver above. AI provenance for the development is tracked with `ailog`. Full quantitative results are tabulated in appendix A.

A Validation data

All numbers are from the large-scale sweep (`FAST128` release; operator runs use the Arnoldi solver of §2). Run parameters per experiment are in table 1. Uniform character-sector gaps are reported as $\kappa = 1 - \max_\chi[\rho(\mathcal{L}_\chi)/\rho(\mathcal{L}_{\mathbf{1}})]$ (the worst character; an earlier draft's $1 - \min$ figures are superseded).

A.1 Experiment B — Cramér-tilt bulk (octave 26, middle segments, $\ell \geq 16$, 1.04×10^5 segs)

	$\hat{p}_1$	$\hat{p}_2$	$\hat{p}_3$	$\hat{p}_4$	entropy/digit
empirical	0.657	0.229	0.089	0.021	0.870
annealed $p_b = 2^{-b}$	0.500	0.250	0.125	0.0625	—
critical $p_b^{(1)} = 3 \cdot 4^{-b}$	0.750	0.1875	0.0469	0.0117	0.811

Table 1: Run parameters and headline result, by experiment.

Exp	parameters	headline result
H	seeds 3–10^6, $r \le 8$; GMP spot-check	23,905,968/23,905,968 match; 144/144 GMP
A	octaves 10–28, 10^5 seeds/oct	tail slope ≈ -0.97
B	octave 26, 2×10^5 seeds, `height`	middle bulk leans to $p^{(1)}$ (below)
C	50 long-orbit seeds; $Q \in \{9, 27, 81\}$	$\lvert W\rvert$ decays with window length
D	octave 24, 10^5 seeds	$M_{\mathrm{post}}(1) = 0.989 < 1$
E	$Q \le 243$, L=48, R=6, $B_{\max}$=20, full group	conductor table (below)
F	$Q \le 2187$, $B_{\max}$=28, full group	κ_Q stabilises (below)
G	$Q \in \{9, 27\}$, small L	no spurious peripheral structure
I	octaves 18–24, 8×10^4/oct	1,254,232 windows, 4.2% flagged

$D\big(\hat{p} \,\big\|\, p^{(1)}\big) = 0.093 \ll D(\hat{p} \,\|\, p) = 0.191$. (Provisional: depends on the working criticality label.) *Exp. D* (provisional): 6,496 post-exit windows, $D(\hat{p}_{\mathrm{post}} \,\|\, p) = 0.177 \ll D\big(\hat{p}_{\mathrm{post}} \,\big\|\, p^{(1)}\big) = 0.576$, mean $M_{\mathrm{post}}(1) = 0.989$. *Exp. C*: $|W|(s{=}0, Q{=}27, k{=}1)$ by window length $L = 16, 32, 77, 116$: $0.31, 0.24, 0.20, 0.11$.

A.2 Experiments E/F — operator gaps

Residue-only uniform gap $\kappa_Q = 1 - \max_\chi$ ratio (full character group, $k = 1..\lfloor \phi/2 \rfloor$):

Q	9	27	81	243	729	2187
κ_Q (s=0)	0.345	0.236	0.236	0.236	0.236	0.236
κ_Q (s=1)	0.092	0.092	0.074	0.074	0.074	0.074

The gap stabilises (0.236/0.074); the worst character sits at a fixed frequency, not at $k = 1$. Annealed vs. exact Fourier (48 sectors): correlation 0.38, mean deviation 0.25; the lowest frequency $k = 1$ is gapped by the exact operator by 0.24–0.78 whereas the annealed $1 - |\hat{p}(1/\phi)| \to 0$.

Faithful `sync` gap by conductor (after the coboundary quotient; each ratio r-independent by conductor autonomy, main paper):

conductor	3 (χ_2)	9	27	81	243
order	2	6	9	54	81
gap ratio (s=0)	1.0000	**0.9069**	0.8530	0.8261	0.7306
κ	0 (cob.)	**0.0931**	0.1470	0.1739	0.2694

Each entry equal across $Q \in \{9, 27, 81, 243\}$; worst non-coboundary sector at the fixed conductor 9, $\kappa_0 \approx 0.0931$. **Diagnostic (`factored`):** the quadratic sector is gapped to 0.333 (s=0)/0.600 (s=1) and κ equals the residue-only value sector-by-sector, confirming χ_2 is the unique difference between faithful and factored.

A.3 Experiment I and the shadow↔operator relation (provisional labels)

by label	windows	frac. shadow
`critical_C_A`	61,372	0.084
`high_ascent`	445,193	0.054
`post_exit`	747,621	0.032

by length	windows	frac. shadow
8–15	331,996	0.003
16–31	330,930	0.029
32–63	591,088	0.072
64–127	218	1.000

Join to residue-only gaps (53,158 flagged windows): 82% at $s = 1$ (100% of long ones); long shadows in weaker-gap sectors (mean ratio 0.73 vs. 0.57); corr$(|W|, \text{gap}) = +0.25$; 94% pick $Q = 9$ (finite-class inflation).

B Addendum: the threshold-Schur and root-ball campaign

The analytic endgame of the main paper (threshold Schur at $m = c$; bounded shells; cascade contraction; class-IV root-ball analysis) was driven by a second suite of audits, archived under `c_suite/analysis/`. For each script we list the headline finding; full provenance is in the `ailog` record (50 captures, validation passing).

- `threshold_class_audit.py` — exact four-class decomposition of the G_c Schur rows at $m = c$ ($c = 4, 5$, full word enumeration): class I machine-zero; twisted near-locked class controls (58%/48%).
- `cascade_row_audit.py` — heavy-row ($x = -1$) cascade audit with the two-point closed form; $\Lambda_c = 1.0392, 1.0051, 0.9827$ ($c = 5, 6, 7$): crossing below 1 at $c = 7$; signed/abs $\to 0.96$; off-representative mass collapsing.
- `cascade_operator.py` — scaled two-channel cascade chain to $c = 30$: bulk per-level Perron radius exactly 3/4; Perron vector a point mass at the locked state; model/measured agreement 0.5%.
- `opflat_levelwise.py` — exact Plancherel shells of the renewal constants ($L \le 7$): primitive shells constant ≈ 1.4; levelwise maximum grows along the dyadic-resonant direction $u/3^L \to 3/4$ (L^∞ false, L^2 true).
- `degmin_mass_audit.py`, `degmin_validate.py` — degenerate-minimum mass law; classifier validated against exact sums (machine zeros vs $|S|/\varphi \in \{1/2, 1\}$ exactly).
- `classIV_levelcost.py` — level-cost law per nominal shell: $\Pr[h \ge 1 \mid d] \approx 0.13$, conditionals $0.30, 0.33$; first level a-independent without exception; singleton shells cannot degenerate (8.9×10^5 pairs, zero events).
- `cellsum_gauss_audit.py` — quadratic valuation $v_3(\mathcal{L}') = d + i_{\min}$ on 300/300 class-IV points.
- `descent_cancellation.py` — no phase cancellation on degenerate pairs ($|S_{\text{deep}}|/m_{\text{deep}} \approx 0.9$–$1.0$): their control is purely probabilistic.
- `midstrata_c9.py` — the decisive $c = 9$ spectrum: Fubini density $0.3 \cdot 3^{-h}$ with successive ratios exactly $\frac{1}{3}$; the root-ball law.

Headline analytic constants established in the main paper: pinning $C_0 = \prod_j (1 - 2^{-2\cdot 3^j})^{-1} < 1.355$; shell bound $\tilde{Q}_L \leq 3C_1 < 5.16$, $C_1 = \prod_j \frac{1+2^{-2\cdot 3^j}}{1-2^{-2\cdot 3^j}}$; level-cost $\lambda = \frac{4}{7} < 3^{-1/2}$; cascade and class-IV row rates $(\sqrt{3}/2)^c$; asymptotic gap $1 - (\sqrt{3}/2)^{1/2} \approx 0.069$ at $s = 0$.

References

[1] W. E. Arnoldi, *The principle of minimized iterations in the solution of the matrix eigenvalue problem*, Quart. Appl. Math. **9** (1951), 17–29.

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[4] Y. Saad, *Numerical Methods for Large Eigenvalue Problems*, 2nd ed., Classics in Applied Mathematics **66**, SIAM, Philadelphia, 2011.